

Magnetism and Matter

SHORT NOTES

- ❖ A natural magnet is an ore of iron (Fe_3O_4) which attracts small pieces of iron, cobalt and nickel towards it.
- ❖ Lode stone is a natural magnet.
- ❖ Unit pole is defined as that pole which when placed in vacuum (or in air) at a distance of one meter from an equal and similar pole, repels it with a force equal to 10^{-7} newton.
- ❖ A current carrying loop behaves as a magnet *i.e.*, magnetic dipole. Thus magnetic dipole moment of a current loop *i.e.*, $M = IA$ where, A = area of the loop.
- ❖ A current carrying straight solenoid behaves like a bar magnet.
- ❖ Isoclinic line corresponding to zero dip is called **aclinic line**, or **magnetic equator**.
- ❖ A line passing through places having equal values of B_H is called **isodynamic line**.

- ❖ **Oscillation magnetometer** is used for comparing magnetic moments of two magnets and also for comparing the horizontal component of earth's field at two places.
- ❖ In a vibration magnetometer, the period of oscillation T is given by $T = 2\pi\sqrt{I / MB_H}$, where I is the moment of inertia of the magnet about the suspension.
- ❖ Frequency of oscillation in a oscillation magnetometer is given by, $n = \frac{1}{2\pi}\sqrt{MB_H / I}$.
- ❖ If two magnets are placed one above the other symmetrically and allowed to oscillate with a period, T_1 in a horizontal plane with a uniform field and with a period T_2 when one of the magnets is reversed, then $\frac{M_1}{M_2} = \frac{T_2^2 + T_1^2}{T_2^2 - T_1^2}$, M_1 and M_2 being the moments of the magnets.

Magnetic quantities-Units and dimensions

S.No.	Quantity	S.I. Unit	Dimensions
1.	Magnetic moment (vector quantity), $M = IA$, $M = m \times l$	Am^2	L^2A^1
2.	Pole strength (scalar), $m = M/L$	Am	LA
3.	Intensity of magnetisation (vector) $I = M/V$	Am^{-1}	$L^{-1}A$
4.	Magnetic flux (Scalar) $\phi = BA$	weber	$ML^2T^{-2}A^{-1}$
5.	Magnetic induction (vector) B	Wb m^{-2} or tesla Am^{-1}	$MT^{-2}A^{-1}$
6.	Intensity of magnetic field (vector)	Am^{-1}	$L^{-1}A$
7.	Magnetic permeability (scalar), $\mu = B/H$	H/m	$MLT^{-2}A^{-2}$
8.	Relative permeability (scalar), $\mu_r = \mu/\mu_0$	unitless	zero dimension
9.	Magnetic susceptibility (scalar) $\chi_m = I/H$, $\chi_m = (\mu_r - 1)$	unitless	zero dimension
10.	Periodic time of a magnet (scalar) $T = 2\pi\sqrt{\frac{I}{MB_H}}$ (I = moment of inertia)	sec	T

- ❖ **Electromagnets:** The material for cores of electromagnets should have maximum flux density with comparatively small magnetising field and low hysteresis loss. Soft iron is best suited for electromagnet. The hysteresis loop is thin and long. Due to the small area of hysteresis loop, energy loss is small.
- ❖ For soft iron retentivity/ramnant magnetism is high, coercivity is low, magnetic permeability μ is high and magnetic susceptibility (χ_m) is high.
- ❖ For steel, ramnant magnetism is low, coercivity is high, μ is low and χ_m is low.
- ❖ About 90% of magnetic moment is due to spin motion of electrons while remaining 10% is due to their orbital motion.
- ❖ When a current I flows in a coil of effective area A and number of turns N , the magnetic moment of coil, $M = NIA$. where, A = area of coil = πr^2 , $M = N\pi r^2$.
- ❖ The magnetic moment of a current carrying solenoid = NIA . M acts along the axis of solenoid.
- ❖ All substances exhibit diamagnetism. In paramagnetic and ferromagnetic substances, the diamagnetism is neutralised by the large intrinsic dipole moment of spinning electrons.
- ❖ Magnetic length = $\frac{5}{6}$ × geometric length of magnet.
- ❖ The perpendicular bisector of magnetic axis is known as neutral axis of magnet. Magnetism at neutral axis is zero and at poles is maximum.